Unit 2

Lesson Four

Cubic roots of unity

* We know that the trigonometric form of the number 1 is $\cos 0 + i \sin 0$

* If $z = 1$

Then the cubic roots of the number z are got from the relation :

$z^{\frac{1}{3}} = (\cos 0 + i \sin 0)^{\frac{1}{3}}$, by using De moivre's theorem

$\therefore z^{\frac{1}{3}} = \left(\cos \frac{2\pi k}{3} + i \sin \frac{2\pi k}{3}\right)$ where $k = 0 , 1 , 2$

, at $k = 0$ $\quad \therefore z^{\frac{1}{3}} = \cos 0° + i \sin 0° = 1$

, at $k = 1$ $\quad \therefore z^{\frac{1}{3}} = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = \frac{-1}{2} + \frac{\sqrt{3}}{2} i$

, at $k = 2$ $\quad \therefore z^{\frac{1}{3}} = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} = \frac{-1}{2} - \frac{\sqrt{3}}{2} i$

i.e. The cubic roots of unity are $1 , \frac{-1}{2} + \frac{\sqrt{3}}{2} i , \frac{-1}{2} - \frac{\sqrt{3}}{2} i$

Notice that :

We can get the cubic roots of unity by solving the equation $z^3 = 1$:

$z^3 - 1 = 0$ $\quad \therefore (z - 1)(z^2 + z + 1) = 0$

at $z - 1 = 0$ $\quad \therefore z = 1$

and at $z^2 + z + 1 = 0$ (by using the formula) $a = 1$, $b = 1$, $c = 1$

$\therefore z = \frac{-1 \pm \sqrt{-3}}{2} = \frac{-1 \pm \sqrt{3} i}{2}$ $\quad \therefore$ The cubic roots are $1 , \frac{-1}{2} + \frac{\sqrt{3}}{2} i , \frac{-1}{2} - \frac{\sqrt{3}}{2} i$

Remark

The cubic roots of the number one are complex numbers one of them is real "1"

and the other roots are conjugate complex numbers , square of one of them is equal to the other.

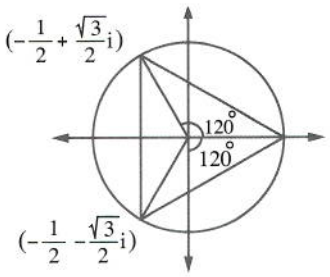
i.e. $\left(\frac{-1}{2} + \frac{\sqrt{3}}{2} i\right)^2 = \frac{-1}{2} - \frac{\sqrt{3}}{2} i$, $\left(\frac{-1}{2} - \frac{\sqrt{3}}{2} i\right)^2 = \frac{-1}{2} + \frac{\sqrt{3}}{2} i$

We can symbolized of one of then by ω (Omiga) , then the other is ω^2

Then : The cubic roots of one are $1 , \omega , \omega^2$

Properties of the cubic roots of one :

If 1 , ω , ω^2 are the cubic roots of unity :



1. The cubic root of unity represented by points of Argand diagram lying on a circle its centre is the origin point and its radius = 1 unit and represent the vertices of equilateral triangle.

2. The sum of the cubic roots of unity = zero **i.e.** $1 + \omega + \omega^2 = 0$

Proof :

$$1 + \left(\frac{-1}{2} + \frac{\sqrt{3}}{2}\,i\right) + \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}\,i\right) = 1 + \frac{-1}{2} + \frac{\sqrt{3}}{2}\,i + \frac{-1}{2} - \frac{\sqrt{3}}{2}\,i = \text{zero}$$

Another Proof :

$\because$ ω , ω^2 are roots of the equation $z^2 + z + 1 = 0$

which resulting from solving the equation $z^3 = 1$

$\therefore$ Sum of roots : $\omega + \omega^2 = \dfrac{-\text{ coeff. of } z}{\text{coeff. of } z^2} = \dfrac{-1}{1} = -1$

$\therefore$ $1 + \omega + \omega^2 = 1 + (-1) = \text{zero}$

Remark

$\because$ $1 + \omega + \omega^2 = 0$

$\omega + \omega^2 = -1$	$1 + \omega = -\omega^2$	$1 + \omega^2 = -\omega$

3. Product of the two not real cubic roots of unity = 1

$\therefore$ $\omega \times \omega^2 = 1$ $\therefore$ $\omega^3 = 1$

Proof :

$$\left(\frac{-1}{2} + \frac{\sqrt{3}}{2}\,i\right) \times \left(\frac{-1}{2} - \frac{\sqrt{3}}{2}\,i\right) = \frac{1}{4} + \frac{3}{4} = 1$$

$\therefore$ $\omega \times \omega^2 = 1$ $\therefore$ $\omega^3 = 1$

Another Proof :

$\because$ ω , ω^2 are roots of the equation $z^2 + z + 1 = 0$

$\therefore$ Product of them $= \dfrac{\text{Absolute term}}{\text{coeff. of } z^2} = \dfrac{1}{1} = 1$

i.e. $\omega^3 = 1$

4. The difference between the two not real cubic roots of unity $= \pm\sqrt{3}\,i$

i.e. $\omega - \omega^2 = \pm\sqrt{3}\,i$ or $\omega^2 - \omega = \pm\sqrt{3}\,i$

Proof :

$\because (\omega - \omega^2)^2 = (\omega^2 - \omega)^2 = \omega^4 - 2\,\omega^3 + \omega^2 = \omega^3 \times \omega - 2 \times 1 + \omega^2$

$= \omega + \omega^2 - 2 = -1 - 2 = -3 = \omega + \omega^2 - 2 = -1 - 2 = -3$

$\therefore \omega - \omega^2 = \pm\sqrt{-3} = \pm\sqrt{3}\,i$ $\quad\quad$ $\therefore \omega^2 - \omega = \pm\sqrt{-3} = \pm\sqrt{3}\,i$

Remark

If $z = a + b\,\omega$, then $\bar{z} = a + b\,\omega^2$ where $a \in \mathbb{R}$, $b \in \mathbb{R}$

The integer power of ω

$\because \omega$ is one of cubic roots of the equation $z^3 = 1$ $\quad\quad$ $\therefore \omega^3 = 1$

Generally for all $n \in z$:

1. $\omega^{3n} = 1$ $\quad\quad$ **i.e.** $\omega^3 = \omega^6 = \omega^9 = \cdots = 1$

2. $\omega^{3n+1} = \omega$ $\quad\quad$ **i.e.** $\omega^4 = \omega^7 = \omega^{10} = \cdots = \omega$

3. $\omega^{3n+2} = \omega^2$ $\quad\quad$ **i.e.** $\omega^5 = \omega^8 = \omega^{11} = \cdots = \omega^2$

and we can notice that :

- $\omega^{-1} = \frac{1}{\omega} = \frac{\omega^3}{\omega} = \omega^2$ $\quad\quad$ **i.e.** $\frac{1}{\omega} = \omega^2$
- $\omega^{-2} = \frac{1}{\omega^2} = \frac{\omega^3}{\omega^2} = \omega$ $\quad\quad$ **i.e.** $\frac{1}{\omega^2} = \omega$

For example :

- $\omega^{16} = \omega^{15+1} = \omega$
- $\omega^{47} = \omega^{45+2} = \omega^2$
- $\omega^{-25} = \frac{1}{\omega^{25}} = \frac{1}{\omega^{24+1}} = \frac{1}{\omega} = \omega^2$
- $\omega^{-101} = \frac{1}{\omega^{101}} = \frac{1}{\omega^{99+2}} = \frac{1}{\omega^2} = \omega$

Example 1

If 1 , ω , ω^2 are the cubic roots of unity , then prove that :

1. $(1 + \omega)^{16} = \omega^2$

2. $(1 - \omega - \omega^2)^3 = 8$

3. $(1 - \omega)(1 - \omega^2) = 3$

4. $(1 - 2\,\omega + \omega^2) \cdot (1 + \omega - 5\,\omega^2) = 18$

Solution

1. $\because 1 + \omega = -\omega^2$

$\therefore (1 + \omega)^{16} = (-\omega^2)^{16}$

$\therefore (1 + \omega)^{16} = \omega^{32} = \omega^{30+2} = \omega^2$

2. $1 - \omega - \omega^2 = 1 - (\omega + \omega^2) = 1 - (-1) = 2$

$\therefore (1 - \omega - \omega^2)^3 = (2)^3 = 8$

3. $(1 - \omega)(1 - \omega^2) = 1 - \omega - \omega^2 + \omega^3$

$= 1 - (\omega + \omega^2) + 1 = 1 - (-1) + 1 = 1 + 1 + 1 = 3$

4. $\because 1 - 2\omega + \omega^2 = (1 + \omega^2) - 2\omega = -\omega - 2\omega = -3\omega$

, $1 + \omega - 5\omega^2 = (1 + \omega) - 5\omega^2 = -\omega^2 - 5\omega^2 = -6\omega^2$

$\therefore (1 - 2\omega + \omega^2)(1 + \omega - 5\omega^2) = -3\omega \times -6\omega^2 = 18\omega^3 = 18$

Example 2

Prove that : $\left(\frac{1}{1+2\omega} - \frac{1}{1+2\omega^2}\right)^2 = \frac{-4}{3}$

Solution

$$\frac{1}{1+2\omega} - \frac{1}{1+2\omega^2} = \frac{(1+2\omega^2) - (1+2\omega)}{(1+2\omega)(1+2\omega^2)} = \frac{2\omega^2 - 2\omega}{1+2\omega+2\omega^2+4\omega^3}$$

$$= \frac{2(\omega^2 - \omega)}{1+2(\omega^2+\omega)+4} = \frac{2(\omega^2-\omega)}{1-2+4} = \frac{2}{3}(\omega^2 - \omega)$$

$$\therefore \left(\frac{1}{1+2\omega} - \frac{1}{1+2\omega^2}\right)^2 = \frac{4}{9}(\omega^2-\omega)^2 = \frac{4}{9}\left(\pm\sqrt{3}\,i\right)^2 = \frac{4}{9} \times -3 = \frac{-4}{3}$$

Example 3

Find the value : $\left(\frac{2+3\omega}{2\omega^2+3} + \frac{5\omega^2-3}{5-3\omega}\right)^{15}$

Solution

$$\because \frac{2+3\omega}{2\omega^2+3} = \frac{2\omega^3+3\omega}{2\omega^2+3} = \frac{\omega(2\omega^2+3)}{2\omega^2+3} = \omega$$

$$\frac{5\omega^2-3}{5-3\omega} = \frac{5\omega^2-3\omega^3}{5-3\omega} = \frac{\omega^2(5-3\omega)}{(5-3\omega)} = \omega^2$$

$\therefore$ The value $= (\omega + \omega^2)^{15} = (-1)^{15} = -1$

Example 4

Form the quadratic equation whose roots are : $\frac{2+\omega}{1-\omega}$, $\frac{2+\omega^2}{1-\omega^2}$

Solution

Sum of the roots $= \frac{2+\omega}{1-\omega} + \frac{2+\omega^2}{1-\omega^2} = \frac{(2+\omega)(1-\omega^2)+(2+\omega^2)(1-\omega)}{(1-\omega)(1-\omega^2)}$

$= \frac{2-2\omega^2+\omega-\omega^3+2-2\omega+\omega^2-\omega^3}{1-\omega^2-\omega+\omega^3}$

$= \frac{2-2\omega^2+\omega-1+2-2\omega+\omega^2-1}{1-\omega^2-\omega+1} = \frac{2-(\omega+\omega^2)}{2-(\omega+\omega^2)} = \frac{3}{3} = 1$

, product of the roots $= \frac{2+\omega}{1-\omega} \times \frac{2+\omega^2}{1-\omega^2} = \frac{4+2(\omega+\omega^2)+\omega^3}{1-(\omega+\omega^2)+\omega^3} = \frac{4+2(-1)+1)}{1-(-1)+1} = \frac{3}{3} = 1$

$\therefore$ The equation is $x^2 - (\text{sum})\, x + \text{product} = 0$

i.e. The equation is $x^2 - x + 1 = 0$

Example 5

Find the solution set in C for each of the following equations :

1. $x^2 - 2\omega x - \omega = 0$ **2.** $x = 1 + \omega^x + \omega^{2x}$

3. $(2x+3)^6 - 9(2x+3)^3 + 8 = 0$

Solution

1. $\because x^2 - 2\omega x - \omega = 0$ $\quad a = 1$, $b = -2\omega$, $c = -\omega$

$\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{2\omega \pm \sqrt{4\omega^2 + 4\omega}}{2}$

$= \frac{2\omega \pm \sqrt{4(\omega^2+\omega)}}{2} = \frac{2\omega \pm 2\sqrt{-1}}{2} = \frac{2\omega \pm 2i}{2} = \omega \pm i$

$\therefore$ S.S. $= \{\omega + i , \omega - i\}$

2. $\because x = 1 + \omega^x + \omega^{2x}$ $\quad \therefore x - 1 = \omega^x + \omega^{2x}$

$\therefore (x-1)^2 = (\omega^x + \omega^{2x})^2$

$\therefore x^2 - 2x + 1 = \omega^{2x} + 2\omega^{3x} + \omega^{4x} = \omega^{2x} + 2 + \omega^x$

$\therefore x^2 - 2x - 1 = \omega^{2x} + \omega^x$

$\because \omega^{2x} + \omega^x = x - 1$ $\quad \therefore x^2 - 2x - 1 = x - 1$

$\therefore x^2 - 3x = 0$ $\quad \therefore x(x-3) = 0$

$\therefore x = 0$ (does not satisfy) or $x = 3$ $\quad \therefore$ S.S. $= \{3\}$

3. $(2x+3)^6 - 9(2x+3)^3 + 8 = 0$

$\therefore (2x+3)^3 = 1$

$\therefore x = -1$ or $2x+3 = \omega$

$\therefore x = \dfrac{\omega^2 - 3}{2}$

Or $(2x+3)^3 = 8$

$\therefore x = \dfrac{-1}{2}$ or $2x+3 = 2\omega$

$\therefore x = \dfrac{2\omega^2 - 3}{2}$

$\therefore$ S.S. $= \left\{-1, \dfrac{\omega-3}{2}, \dfrac{\omega^2-3}{2}, \dfrac{-1}{2}, \dfrac{2\omega-3}{2}, \dfrac{2\omega^2-3}{2}\right\}$

$\therefore \left((2x+3)^3 - 1\right)\left((2x+3)^3 - 8\right) = 0$

$\therefore 2x+3 = 1$

$\therefore x = \dfrac{\omega-3}{2}$ or $2x+3 = \omega^2$

$\therefore x = \dfrac{2\omega-3}{2}$ or $2x+3 = 2\omega^2$

Example 6

If $(1-i)(x+yi) = 19\left(\dfrac{1}{2-3\omega^2} + \dfrac{1}{5+3\omega^2}\right)$ then find the value of : x , y

Solution

$$(1-i)(x+yi) = 19\left(\frac{5+3\omega^2+2-3\omega^2}{(2-3\omega^2)(5+3\omega^2)}\right) = 19\left(\frac{7}{10+6\omega^2-15\omega^2-9\omega^4}\right)$$

$$= 19\left(\frac{7}{10-9\omega^2-9\omega}\right) = 19\left(\frac{7}{10-9(\omega^2+\omega)}\right) = 19\left(\frac{7}{10+9}\right) = 7$$

$$\therefore x+yi = \frac{7}{1-i} \times \frac{1+i}{1+i} = \frac{7+7i}{1+1} = \frac{7}{2} + \frac{7}{2}i$$

$$\therefore x = \frac{7}{2} \ , \ y = \frac{7}{2}$$

Example 7

Find value of : $\sum_{r=1}^{20} (-1)^{r+1} \times \omega^r$

Solution

$$\sum_{r=1}^{20} (-1)^{r+1}\omega^r = (-1)^2\omega + (-1)^3\omega^2 + (-1)^4\omega^3 + \ldots + (-1)^{21}(\omega^{20})$$

$$= \omega - \omega^2 + \omega^3 \ldots - \omega^{20}$$

It is the sum of geometric sequence its 1st term is ω and its base is $\dfrac{-\omega^2}{\omega} = -\omega$ and the number of its terms is 20

$$\therefore S_n = \frac{a(r^n - 1)}{r-1} \ , \ S_{20} = \frac{\omega((-\omega)^{20} - 1)}{-\omega - 1}$$

$$S_{20} = \frac{\omega(\omega^{20} - 1)}{-(\omega+1)} = \frac{\omega^{21} - \omega}{\omega^2} = \omega^{19} - \frac{1}{\omega} = \omega - \omega^2 = \pm\sqrt{3}\,i$$

$$\sum_{r=1}^{20} (-1)^{r+1}\omega^r = \pm\sqrt{3}\,i$$